

2017 H2 Q3

Section: Our Dynamic Universe

Topic: Vertical motion under gravity

A ball is thrown vertically upwards at 5.6 m s^{-1} from above the ground. Ignore air resistance. Graph shows vertical velocity vs time until just before impact (final $v \approx -7.7 \text{ m s}^{-1}$).

(a)(i) Time to reach maximum height

At max height $v = 0$. Equation: $v = u + at$.

$$0 = 5.6 - 9.8 t \Rightarrow t = 5.6 / 9.8 = 0.571 \text{ s}.$$

Answer: $\approx 0.57 \text{ s}$

(a)(ii) Distance fallen from maximum height to ground

Use $v^2 = u^2 + 2as$ from max height ($u = 0$).

$$v^2 = 0^2 + 2(-9.8)s \Rightarrow s = v^2/(2g).$$

$$s = (7.7^2)/(2 \times 9.8) = 29.4/19.6 = 3.03 \text{ m}.$$

Answer: $\approx 3.0 \text{ m}$

(b) Graph shape for greater initial velocity

The velocity-time line would:

- start higher on the v -axis (greater positive intercept),
- still have slope -9.8 m s^{-2} ,

- cross the t-axis later (longer time to max height),
- extend further into negative region (greater downward velocity before impact).

Revision tips

- Time to maximum height: $t = u/g$.
- Displacement from max to ground: use $v^2 = u^2 + 2as$ with $u = 0$.
- On velocity-time graphs, slope = acceleration = -9.8 m s^{-2} always.
- Greater initial speed shifts line upwards and lengthens total flight time.